

Discussion 8: Symbolic planning and Neural Networks

Discussion Topic:

Symbolic planning systems, including robotic control and any form of cognitive AI system, from DSS expert systems to automated logistics control software, will need to leverage connectionist techniques alongside symbolic knowledge representation systems. How might integrating connectionist methods (e.g., neural networks) into symbol-based planning systems improve explainability?

My Post:

Hello Class

I think integrating connectionist methods with symbolic planning can improve explainability, but the improvement depends on how the two approaches are connected. Adding an Artificial Neural Network (ANN) does not automatically make the system's reasoning easier to understand. In fact, it can add less interpretability and may make the reasoning system itself a black box. In other words, symbolic systems use concepts and relationships people can understand to represent rules; however, by definition, the nature of the learned internal representations of ANNs, which are often distributed across many weights and activations, can infiltrate concepts and relationships within the symbolic system in ways that are difficult to describe or interpret. Nonetheless, there is still value in combining symbolic planning with an ANN. The ANN can turn observations in the data into recognizable conditions, which can then be integrated into the symbolic system and used by the planner to develop a symbolic plan.

In practice:

Data -> ANN identifies a condition -> condition enters the symbolic system -> symbolic planner uses it to develop the plan.

This makes it easier to understand what the system has recognized in the data and how that information influenced the symbolic planner's choice of actions.

Gordon et al. (2019) provide a good example of how this connection can work through their Hierarchical Planning and Reinforcement Learning (HIP-RL) system. In HIP-RL, neural perception components identify objects in the environment and use those observations to update a shared knowledge state. The symbolic planner can then use that knowledge to determine what actions are possible based on their preconditions and effects. A precondition is simply something that must already be true before an action can happen. For example, if a robot needs to retrieve an object from a closed drawer, the drawer must first be opened. I think this is where the explainability benefit becomes clearer. The system could show what the ANN detected, how that information became part of the symbolic knowledge, and how the planner used it when selecting the next action.

For example, a warehouse robot equipped with a visual object detection system may use a symbolic neural vision model, for instance, to detect a pallet blocking an aisle. The detection produced by the visual model could then be translated into a condition within the symbolic system, such as the aisle being blocked. The symbolic planner could use that condition together with its existing rules to decide that the aisle should not be used, even if it would normally provide the shortest route. In this example, I would want the explanation to show what the vision model detected, how that detection became a symbolic condition, which rule was applied to that condition, and why the planner selected a different route. This would make it easier for an operator to trace an incorrect decision back to its source. For instance, the

visual model may have incorrectly identified the aisle as blocked, or the detection may have been correct while the symbolic rule caused the planner to make an unnecessary decision.

Learning from experience could also become part of this process. Yang et al. (2018) describe a system in which symbolic plans guide the agent's actions and learning, while information learned during execution is fed back into the symbolic knowledge and used to improve future planning. Lyu et al. (2019) take a similar approach with deep reinforcement learning by connecting symbolic actions with learned subtasks, which they describe as task-level interpretability. In the warehouse example, the robot could learn through repeated experience that one route takes longer, is less reliable, or creates more execution problems than another route. That learned information could then become part of the symbolic planning process. If the system keeps a clear connection between what was learned and how it affected the symbolic plan, an operator could see both the rule that influenced the decision and the learned information that caused the planner to prefer one route over another.

However, this also shows an important limitation of combining ANNs with symbolic planning. The symbolic planner can reason correctly and still produce a poor plan if the information it receives from the ANN is wrong. In the warehouse example, the vision model could incorrectly identify a pallet as blocking an aisle. The symbolic system may then treat that prediction as a valid condition and avoid the aisle, even though nothing is actually blocking it. For this reason, I think the system should keep uncertain ANN predictions separate from verified facts and allow an operator to review or correct them before replanning. This is where explainability becomes especially important because the user should be able to trace the decision from the original ANN prediction, through the symbolic condition and rule, to the plan that was ultimately produced.

-Alex

References:

Gordon, D., Fox, D., & Farhadi, A. (2019). *What should I do now? Marrying reinforcement learning and symbolic planning* [Preprint]. arXiv. <https://doi.org/10.48550/arXiv.1901.01492>

Lyu, D., Yang, F., Liu, B., & Gustafson, S. (2019). SDRL: Interpretable and data-efficient deep reinforcement learning leveraging symbolic planning. *Proceedings of the AAAI Conference on Artificial Intelligence*, 33(1), 2970–2977. <https://doi.org/10.1609/aaai.v33i01.33012970>

Yang, F., Lyu, D., Liu, B., & Gustafson, S. (2018). PEORL: Integrating symbolic planning and hierarchical reinforcement learning for robust decision-making. In J. Lang (Ed.), *Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence* (pp. 4860–4866). International Joint Conferences on Artificial Intelligence. <https://doi.org/10.24963/ijcai.2018/675>